2025-2026
MATH
Paper 1

DELIA MEMORIAL SCHOOL (GLEE PATH)

FIRST TERM EXAMINATION 2025-2026

S3 MATHEMATICS
PAPER 1
Question-Answer Book

Time allowed: 1 hour 15 minutes

Group No.	
Name	
Class	**S3 ( )**
Class No.	**( )**
Date	**12 / 01 / 2026**

INSTRUCTIONS

1. Write your group number, name, class and class number on the cover of this answer book.
2. This paper consists of THREE sections, A(1), A(2) and B.
3. Answer ALL questions. Write your answers in the spaces provided in this Question-Answer Book.
4. Graph paper and supplementary answer sheets will be supplied on request.
5. Unless otherwise specified, all working must be clearly shown.
6. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures.
7. The diagrams in this paper are not necessarily drawn to scale.
8. You are required to use **Blue** or **Black PEN** to answer the questions. All **drawings,** if any**,** must be drawn in **sharp HB pencil**.

Section	Marks	
A(1)		30
A(2)		30
B		10
pp -1		
u - 1		
Total		70

OVERALL (Paper 1 & Paper 2)		
Total		100

Section A (1) (30 marks)

1. Factorize $5y^2 - 3x + 5xy - 3y$. (3 marks)

2. Simplify $\frac{(2a)^4}{5a^{-2}}$ and express the answer with a positive index. (3 marks)

3. Evaluate $6.2 \times 10^7 - 4.51 \times 10^7$ without using a calculator and express the answer in scientific notation. (2 marks)

4. A letter is randomly chosen from each of the two words 'SIX' and 'SEVEN'.
   (a) List all the possible outcomes in a table. (1 mark)

1st letter \ 2nd letter	S	E	V	E	N
S	SS				
I					
X					

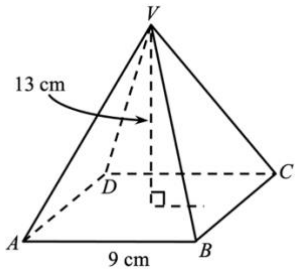
   (b) Find the probability that the two letters chosen are both vowels. (1 mark)

5. The sales amount of a shop was $120 000 in May. Then the sales amount decreased by 10% in June and increased by 10% in July. Find the sales amount of the shop in July. (2 marks)

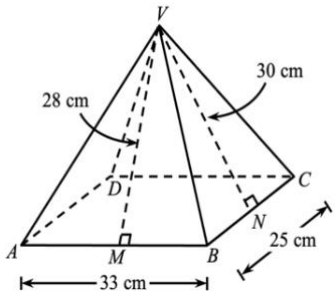
6. Minnie deposits $246 000 in a bank at an interest rate of 0.5% per annum for 18 months on simple interest basis. Find the amount that she will get. (3 marks)

7. The figure shows a right pyramid $VABCD$ of height 13 cm. Its base is a square of side 9 cm. Find the volume of the pyramid.

(2 marks)

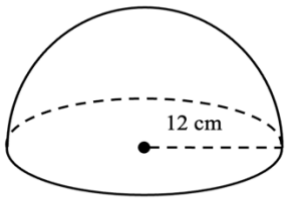


8. The figure shows a right pyramid $VABCD$. Its base is a rectangle of length 33 cm and width 25 cm. The height VM of $\triangle VAB$ is 28 cm while the height VN of $\triangle VBC$ is 30 cm. Find the total surface area of the pyramid. (2 marks)



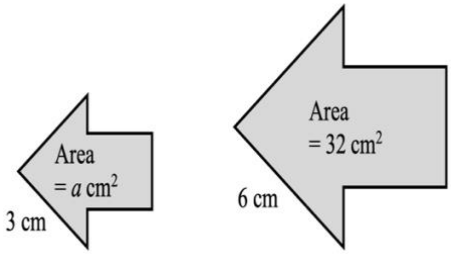
9. The figure shows a hemisphere of radius 12 cm. Express the volume of the hemisphere in terms of π.

(2 marks)



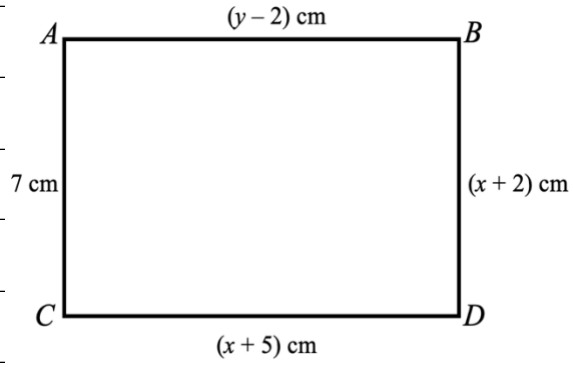
10. Find the unknown in the following pair of similar plane figures.

(2 marks)

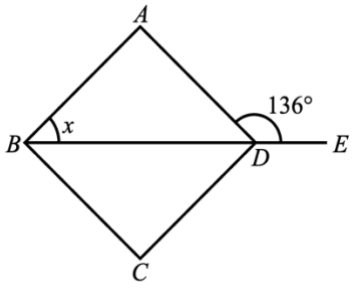


11. In the figure, *ABCD* is a rectangle. Find x and y.

(3 marks)



12. In the figure, *ABCD* is a rhombus. *BDE* is a straight line. $\angle ADE = 136°$. Find x. (4 marks)



Section A (2) (20 marks)

13. (a) Factorize $57st + 27s^2 - 40t^2$. (1 mark)

(b) Hence, factorize $57st + 27s^2 - 40t^2 - 48s - 128t$. (3 marks)

14. Simplify $(x^5y^4)^{-3}\left(\frac{x^6}{y^7}\right)^{-2}$ and express the answer with positive indices. (3 marks)

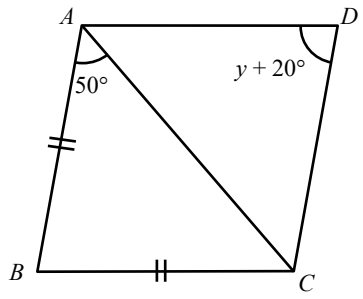
15. A sum of $240 000 is deposited at an interest rate of 4% per annum for 3 years, compounded yearly.

(a) Find the amount. (2 marks)

(b) Find the interest earned in the third year. (3 marks)

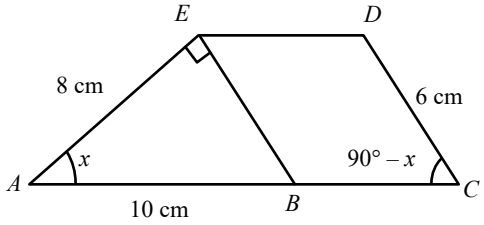
16. In the figure, $ABCD$ is a parallelogram. $AB = BC$ and $\angle BAC = 50°$. Find y.

(3 marks)

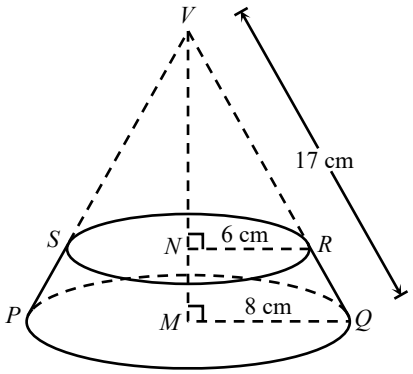


17. In the figure, $ACDE$ is a quadrilateral. B is a point on AC such that $BE \perp AE$. $AB = 10$ cm, $AE = 8$ cm and $CD = 6$ cm. $\angle EAB = x$ and $\angle DCB = 90° - x$. Prove that $BCDE$ is a parallelogram.

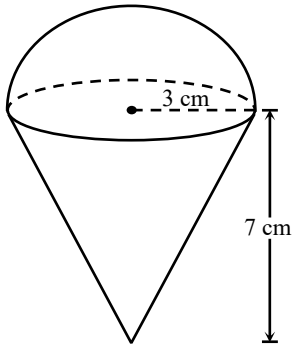
(5 marks)



18. The figure shows a right frustum *PQRS*. The radii of the upper base and the lower base are 6 cm and 8 cm respectively. The slant height of cone *VPQ* is 17 cm. Find the volume of the frustum in terms of π.
(5 marks)



19. The figure shows an ice cream-shaped model which consists of a hemisphere and an inverted right circular cone. The base radius and the height of the circular cone are 3 cm and 7 cm respectively. Find the total surface area of the model. (5 marks)



Section B (10 marks)

20. A solid metal right prism of base area 140 cm^2 and height h cm is melted and recast into two similar solid right pyramids. The bases of the two pyramids are squares. The volume of the smaller pyramid is 960 cm^3 and the ratio of the base area of the smaller pyramid to the base area of the larger pyramid is 4 : 9.

(a) Find the ratio of the volume of the smaller pyramid to the volume of the larger pyramid. (2 marks)

(b) Find the height of the prism. (3 marks)

(c) If the height of the smaller pyramid is 20 cm, find the difference between the total surface area of the larger pyramid and the smaller pyramid. (5 marks)

-End of Paper-